Teasoo Tree Management System (TMS)

Summary of capabilities and a test script · Teasoo Consulting · October 2026

1. What the system is

TMS records every tree planted on Teasoo's reforestation projects and every later visit to it, offline in the field, with no data loss. It has three parts:

- **Android app**: used by field officers and administrators. Works fully offline after the first sign-in and sends data when the phone has internet.
- **Web portal** (`tms.teasooconsulting.com`): used for administration, dashboards, reports and exports. Clients can view their project through a secret share link.
- **Cloud server and database**: stores records and photos, enforces who can do what, and keeps an audit log of every change.

Trees have no individual GPS. Each planting year's land survey divides the area into **sectors** and **sub-sectors**, recorded by their corner points, and a tree is located by its sector and sub-sector.

2. Who uses it

Role	What they can do
Field officer	Register and visit trees in the projects and areas assigned to them, take photos, sync, and back up and restore their phone's data.
Administrator	Everything a field officer can do, plus set up projects, areas, vendors, ID formats, fields, users and assignments; resolve conflicts; correct IDs; export data; share projects with clients; and clear data.
Executive	Read-only dashboards and reports. *Teasoo management* sees all projects; *clients and sponsors* see only the projects they are assigned to.
Client by link	No account needed. Opens one project's read-only dashboard, downloads Excel or CSV, and prints or saves it as a PDF.

3. Capabilities

Field work (Android)

- **Simple home:** one large "Start field session" button. During a session, one large "Register trees" button and a red "End session" button. Large text and buttons.
- **Field session:** choose project, planting year, vendor (if the year has vendors), sector and sub-sector once. Change sub-sector from a drop-down as you move.
- **Register trees:** type only the tree number. The full ID is built from the project's format; numbers run on per project and year, shared by all vendors, and the next one is suggested. Species are chosen from picture cards (the administrator's reference photos; open a card to swipe through all of them and choose). Species and status are remembered. Optional tree height in cm, m, inches or feet. A photo is required, shown whole on the form with Retake and Remove; tap it for full screen. After saving, a confirmation shows the tree's ID before the next form. Date, time and planting date are recorded automatically.
- **Known IDs:** if a tree number is already registered, the app shows the tree's photo, last species and status, and offers "Record a visit".
- **Find a tree:** search by tree number (with photos), open the tree and record a visit: Alive or Dead, a photo, a note and custom fields. The date and time are stamped automatically.
- **Warnings:** the app warns if a tree is visited in a different sector than the one it was registered in.
- **Tree page:** the latest photo at the top, all photos, full visit history and who recorded each visit. Corrections are kept as amendments; nothing is overwritten.
- **Replanting:** plant a replacement for a dead tree. It gets a new ID linked to the old tree; admins can type the ID by hand.
- **My visits:** a list of days with times and counts; open a day to see each tree registered or visited, with its photo.

Offline, sync and safety

- **Saving:** every record is saved on the phone first and sent automatically when there is internet. Sending twice never creates duplicates.
- **Duplicate IDs** entered on two phones are kept as conflicts for an admin to resolve; neither is overwritten.
- **Ready to send:** shows records and photos waiting, with Send now.
- **Sign-out** is blocked while anything is unsent.
- **Photos:** resized to about 300 KB before saving. The phone deletes its copy only after the server confirms it has the photo, and normally keeps a copy for 30 days. **Free up space** removes confirmed copies early.
- **Light backup:** `Download/TMS/TMS-backup.zip` is kept up to date and survives clearing the app or reinstalling it. **Export backup** includes photos; **Restore** puts records back in the queue.
- **Data clear:** when an admin clears the server data, each phone saves a backup, then resets.

Administration (web and Android)

- **Projects:** each has planting years. Every year has its own surveyed sectors and sub-sectors, entered from corner points (walk to a corner and tap "Use my location").
- **Vendors per planting year.**
- **Tree ID format:** one per project, shared by all vendors: fixed parts (project, year, fixed text) and changing parts (sector, sub-sector, tree number). Tree numbers run on per project and planting year. Changes affect new trees only.
- **Custom fields:** extra fields such as a nursery batch number, collected at registration or at every visit. They can be remembered for the rest of the day.
- **Species photos:** several reference photos per species, uploaded on the web or the phone; phones keep them for offline use.
- **Reference lists:** species, statuses and vendors. Anything can be edited, deactivated or deleted (long-press or swipe on the phone, ⋯ on the web). Items still in use are protected.
- **Users:** sign in with a username, phone number or email. Admins can reset passwords and assign people to projects or areas. Each user's activity and last sync are shown.
- **Conflicts and corrections:** resolve duplicate IDs and correct serial numbers. Every change is audited.
- **Activity log:** the latest 10 entries, or the full log filtered by date, person or activity.
- **Clear data:** field data only, or everything. Needs the typed confirmation and your password.

Dashboards, reports and sharing

- **Dashboard:** trees planted, alive and dead, not yet visited, status by sector and species, alive/dead map by area, and weekly field activity.
- **Filters:** project, planting year, vendor, sector, sub-sector, species, status and date range.
- **Visits table:** every visit, newest first, with CSV export; open a row to see its photos.
- **Density map:** each sector and sub-sector drawn from its corner points and shaded by trees per hectare, with alive share and a table (web, Android Administration, and the client link). Trees have no GPS of their own, so density is shown per area.
- **Photos everywhere:** wherever a tree is shown, its photos are too; the latest is shown large on the tree page.
- **Exports:** trees as CSV or Excel. **Export all** gives one Excel workbook containing every trees, visits and configuration sheet. **Print report** saves the dashboard as a PDF.
- **Share with client:** creates a secret link for one project, with an optional expiry, which can be revoked at any time. The client sees a read-only dashboard with Excel, CSV and PDF downloads. Staff names are hidden and each use is recorded.
- **Photo loading:** lists load small thumbnails (about 20 KB); the full photo loads only when opened.

4. Test script

> **Before you start:** an **administrator** account and a **field officer** account (ask Teasoo's administrator). The latest TMS Field app installed on an Android phone (8.0 or newer). A computer browser for the web portal. Tick each step when the expected result is seen; note anything different in the margin.
>
> Test on a test project. Section F clears data: run it only on a system where that is intended.

A. Sign-in and setup (administrator)

#	Do this	Expected result	✔
A1	Open `tms.teasooconsulting.com` and sign in with your username, email or phone number. Try the eye icon on the password field.	The dashboard opens. The eye shows and hides the password.	☐
A2	Projects → New project (name, short code such as `TST`).	The project page opens on the new project.	☐
A3	Add planting year `2026`.	The year appears in the year selector.	☐
A4	Areas → add sector `S01` with 4 corner points, then sub-sector `SS01` inside it.	Outlines are drawn and the sub-sector sits inside the sector. A sub-sector far outside the sector is refused.	☐
A5	Vendors → New vendor (e.g. `Konam`, code `KNM`) and link it to 2026.	The vendor is listed for 2026.	☐
A6	ID format → Change: Project, Year, Tree number; width 5.	The preview shows e.g. `TST-2026-00001`. There is no vendor option.	☐
A6b	Reference data → Add species, and pick 2–3 reference photos in the same form. Use the pencil button on another species to edit it.	The thumbnails appear; clicking one shows it full size.	☐
A7	Custom fields → add "Nursery batch" (text, remember during session).	The field is listed.	☐
A8	Officers → assign the field officer to the project. Set the project status to Active.	The assignment is listed and the project is Active.	☐

B. Field work (Android, field officer)

#	Do this	Expected result	✔
B1	Sign in on the phone with internet on, then wait for the first sync.	The home screen shows a large "Start field session" button; there is no Map button.	☐
B2	Turn on **airplane mode**. Start field session: project, 2026, vendor KNM, S01, SS01.	No "type of visit" or "planted on" is asked. Home shows the session, a large "Register trees" button and a red "End session" button.	☐

B3	Register trees: tap the suggested number, choose the species from its picture card (open its photos once), take a photo, check the whole photo shows (try Retake once), choose Alive, enter a height of 85 cm, add a note (the note box stays above the keyboard), enter a nursery batch, then Save tree.	A "Tree registered" pop-up shows `TST-2026-00001`. "Register next tree" opens an empty form with the batch remembered.	☐
B4	Register 4 more trees. Try saving one without a photo.	Saving is blocked until a photo is taken.	☐
B5	Type an already registered tree number.	"Already registered" appears with the tree's photo, species and status, plus "Record a visit to this tree".	☐
B6	Home → Find a tree → type a number → open the tree → Record a visit → choose Dead, add a photo, Save visit.	The visit is saved. The date and time are stamped automatically (not editable).	☐
B7	Tap outside a text field, and use the hide-keyboard button above the keyboard.	The keyboard closes.	☐
B8	Try to sign out. Later, after syncing, sign out and sign in as the administrator.	Blocked while unsynced: "Sync before signing out". As administrator, the officer's field session is gone (home shows "Start field session").	☐
B9	Open My visits, tap Today; then open Ready to send.	Days are listed with times and counts; Today lists each tree with its photo, marked as waiting. Ready to send shows the counts.	☐
B10	On the phone, open the Files app → Download/TMS.	`TMS-backup.zip` is there.	☐
B11	Turn airplane mode **off**, then Ready to send → Send now.	"Everything is sent". The records show as synced.	☐
B12	Open a tree page.	The latest photo is at the top; photos open full-screen. The history shows who recorded each visit and when.	☐

C. Replanting and corrections (administrator on Android or web)

#	Do this	Expected result	✔
C1	Open the dead tree from B6 → Plant a replacement → switch on "Set ID by hand" and enter `TST-R-0001`, plus a photo.	The new tree is saved and linked to the original. The original keeps its history.	☐
C2	On a synced tree, use Correct serial number with a reason.	The ID changes and the old ID is kept in its history and the activity log.	☐
C3	Correct an earlier visit (wrong status).	A correction is added and the original is marked superseded, not deleted.	☐

D. Dashboards, exports and sharing (web)

#	Do this	Expected result	✔

D1	Dashboard → Filters → choose project, 2026 and vendor KNM.	Counts match the trees entered (alive and dead), and the area map is coloured.	☐
D2	Export all (Excel), then Print report → Save as PDF.	A workbook with Trees, Visits and setup sheets, and a clean one-page-per-section PDF.	☐
D3	Visits → filter by date → open a row → Export CSV.	The row shows that visit's photos. Only matching visits are exported, newest first.	☐
D3b	Web: Density map → choose the project and 2026. Phone: Administration → Density map.	S01 shows one dot per tree (green alive, red dead, grey not visited) scattered inside its outline, with hectares and trees per hectare in the table.	☐
D4	Project → Share → label "Test client", expiry 30 days → Copy. Open the link in a private browser window.	The read-only client dashboard opens without login. Excel, CSV and Print work, and no staff names are shown.	☐
D5	Revoke the link and reload it.	"This link is not valid any more."	☐
D6	Audit log, then See full log → filter by the field officer and today's date.	The latest 10 entries, then that officer's registrations and visits, with device and time.	☐

E. Lists and deletion

#	Do this	Expected result	✔
E1	Add a species that no tree uses, then delete it (⋯ on the web, or long-press or swipe on the phone).	It is deleted, and disappears from phones after their next sync.	☐
E2	Try deleting the species used by your trees, or sector S01.	Refused with a count and the advice to deactivate instead.	☐
E3	Users → create a user with a username only (no email) and sign in with it on the phone.	Sign-in works with the username.	☐

F. Backup, restore and clearing data (only on a test system)

#	Do this	Expected result	✔
F1	Phone: Ready to send → ⋯ → Export backup, then Restore from backup with the same file.	A zip is saved. The restore reports what it re-queued, and nothing is duplicated after sync.	☐
F2	Phone: ⋯ → Free up space.	Space is freed from photos already confirmed by the server. Unsent photos stay.	☐
F3	Web: Settings → Danger zone → Field data only, type `DELETE ALL DATA` and a *wrong* password.	"Your password is incorrect." Nothing is deleted.	☐
F4	Repeat with the correct password. Then sync the phone.	The counts removed are shown. The phone saves a backup, resets, and shows no trees. Projects and users remain.	☐

Report issues with the step number, the phone model, a screenshot and the time it happened (shown in the activity log).